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The sense of agency in perception, behaviour and human–machine interactions

Wen Wen ^{1,2}  and Hiroshi Imamizu ^{1,3,4}

Abstract | The sense of agency refers to the subjective feeling of controlling one's own actions, and through them, external events. The sense of agency is a byproduct of human movements and also greatly shapes perception and behaviour. Furthermore, research on human–machine interaction has highlighted the importance of the sense of agency in joint control between humans and automated systems. In this Review, we first provide an overview of how the sense of agency influences human perception and how the perceptual effects of the sense of agency are used to measure this subjective feeling. Second, we review how the sense of agency modulates behaviour, including action selection, goal-directed actions, and social cognition. Third, we introduce theoretical and neural accounts of how the sense of agency arises. Finally, we explain how the sense of agency applies to human–machine interactions, an area that is rapidly developing and increasingly linked to daily life.

A sense of agency arises when people control their bodies and external events, which leads them to feel responsible for what their actions cause. For example, when someone reaches out to pick up a water bottle, it is unambiguously clear to that person that they performed the action and are the agent who caused the action to occur. A sense of agency can also be felt beyond one's own body. When someone uses buttons to control an avatar in a computer game — although the detailed movements of the avatar are generated by the computer — that person feels a sense of agency over the avatar because its movements match the buttons pressed. However, if a computer glitch occurs, one might temporarily lose the sense of agency owing to the sudden mismatch between the avatar's motion and the buttons pressed. Thus, the sense of agency is linked to a 'who' system, which supports judgements about who is the agent of an action, enabling action awareness and self-consciousness^{1,2}.

Three key aspects underlie the sense of agency: intention, action, and effect. Intention refers to the desired state that one hopes to achieve through their actions. Action refers to the awareness of one's own actions. Finally, effect refers to the perception of the consequences of actions. The processing of actions and effects are directly linked to the motor, perceptual, and sensorimotor systems, each of which has a major role in the sense of agency.

The sense of agency has an important role in shaping perception and behaviour. For example, it shapes where people allocate their attention in the environment^{3–6} and

how intense a stimulus feels⁷. In development, contingency between babies' body movements and visual or auditory feedback of their movements reinforces these movements, increasing the frequency with which they are performed^{8–10}. Thus, the detection of action–effect contingency might have an important role in the development of motor control and the perception of the self¹¹. Agency-related processes also change with age, from childhood to young adulthood to older adulthood^{12–14}. The feeling of control over life events is generally reduced in older adulthood¹² and maintaining an appropriate sense of agency might be important for quality of life in the elderly. In short, the development of the sense of agency starts at a very early age and influences perception and behaviour throughout the lifespan.

The sense of agency is also relevant in the context of psychiatric disorders. Patients with schizophrenia often experience a loss of agency over their thoughts or behaviours^{15–18} and show a poor ability to discern between self-generated and other-generated consequences^{19–22}. Studying the behavioural and neural mechanisms of the sense of agency will provide important knowledge for the diagnosis, understanding, and treatment of these patients.

Finally, there is interest in the sense of agency in engineering because of the rapid development of technologies and systems such as remote control of machines, driving assistance, and rehabilitation robots. These tools aim to reduce human cognitive and physical load and to improve human performance, and they often involve

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Readiness potential

A slow negative electroencephalography (EEG) potential preceding the onset of voluntary movement, associated with movement preparation and decision-making.

human-machine joint control. The sense of agency in such joint control is important in human attention allocation, action selection, and decision-making.

In this Review, we provide an overview of the sense of agency and its influence on perception, behaviour, and human-machine interaction. We first introduce how the sense of agency influences human perception and behaviour. We then discuss the mechanisms and neural bases underlying the sense of agency. Next, we focus on the sense of agency in human-machine interaction. Finally, we highlight new leads for future research into the sense of agency and its applications.

Sense of agency influences perception

Because the sense of agency is a subjective feeling, it can be directly examined through self-report tasks. Yes/no answers to the question ‘Do you feel that you controlled the event?’ and ratings of ‘How much do you feel that you controlled the event?’ are often used. However, despite the rich information contained in self-reports, they can easily be influenced by subjective criteria, bias or disorders in decision-making. For example, a liberal criterion might lead to more ‘yes’ responses compared with a more conservative criterion²³, but the liberal criterion is not necessarily linked to a stronger sense of agency. To deal with these issues, researchers have used indirect (implicit) methods to measure the sense of agency (TABLE 1). These implicit measures involve measuring the difference in perception and behaviour between conditions when people feel a sense of agency versus when they do not, or between conditions when people experience a strong versus a weak sense of agency. Implicit measures are less likely than explicit measures to be affected by individual criteria. In this section, we discuss how the sense of agency influences perception, and the application and limitations of implicit measures. The sense of agency influences the perception of time, the intensity with which perceptual inputs are felt, and how visual attention is directed.

Time perception. The intentional binding effect is an effect of the sense of agency on time perception. The temporal judgement of an action is perceived later if the action causes an effect than if the action does not cause an effect²⁴. Furthermore, an event is perceived to

occur earlier when it is produced by one’s own voluntary action, compared to an event without a preceding action²⁴. In other words, voluntary actions and their effects are bound, resulting in a compressed perception of the time between them. Importantly, the intentional binding effect is not observed when an action is involuntary (for example, when it is triggered by brain stimulation)²⁴.

The original intentional binding paradigm measures action and effect bindings by asking participants to report the timing of a voluntary action and/or the following event^{24,25}. Participants make a temporal judgement about either an action (typically a keypress) onset or an event (typically a tone) onset. Action binding is the difference between the judgements of the action onset for the condition when the keypress action does not cause a tone and for the condition when the keypress action causes a tone (FIG. 1a). Similarly, effect binding is the difference between the judgements of the tone onset for the condition when the tone occurs without a keypress and the condition when the tone is preceded by a keypress (FIG. 1b). Action binding and effect binding reflect different aspects of the intentional binding effect and are useful for separating abnormalities in the perception of actions and sensory feedback^{26–29}. For example, patients with Parkinson’s disease show abnormal action binding, whereas their effect binding is similar to that found in healthy participants^{26,28}, indicating that the disorder of the sense of agency in Parkinson’s disease is linked to a faulty representation of their own actions.

The intentional binding effect and its relationship with the sense of agency have been reproduced in numerous studies. However, the intentional binding effect is sometimes discrepant from the explicit judgement of agency^{30–33}. Some researchers have also argued that the intentional binding effect does not require an intention^{34–37}. For instance, a study using virtual reality showed that passive viewing of a virtual hand pressing a button to trigger a tone produced a binding effect (between the virtual action and the tone) similar to when the tone was triggered by the participants’ actual hand movements³⁷. The observation of the intentional binding effect during passive viewing, without an intention, implies that the intentional binding effect may not be suitable for diagnosing the sense of agency. Furthermore, electrophysiological activity associated with the intentional binding effect, typically measured with electroencephalography (EEG), is not consistent across studies^{38,39}. One study reported that the negative component of the early readiness potential was correlated with the binding effect³⁸. However, another study reported the opposite relationship between readiness potential amplitudes and action binding, showing that the reduced negativity of readiness potential was associated with a stronger binding effect³⁹.

Importantly, many other factors — such as causal relations between events^{34,40,41}, attention^{42,43} and arousal^{33,44} — can affect time perception and influence the binding effect. As a consequence, the intentional binding effect and other implicit measures of the sense of agency should not be used as a diagnosis of the sense

Table 1 | Indirect measurements of the sense of agency

Task	Method	Aspect of sense of agency
Intentional binding	Participants trigger an event by performing a voluntary action and make a judgement regarding the onset of the event and the action.	Temporal perception of voluntary actions and their effects
Sensory attenuation	Participants trigger sensory feedback by performing an action and make a judgement regarding the intensity of the feedback.	Perception of self-induced effects (sensory feedback)
Visual attention	Participants observe a flickering stimulus while continuously controlling (moving) it. Participants detect a visual target among several visual distractors. The level of participant control differs between the target and the distractors.	Attentional allocation to self-induced effect (controllable stimulus)

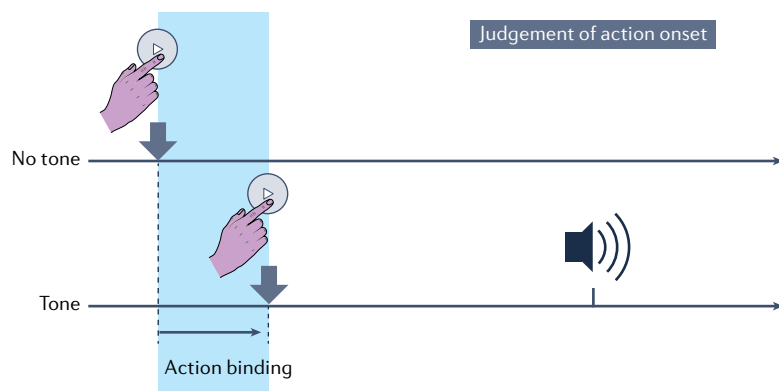
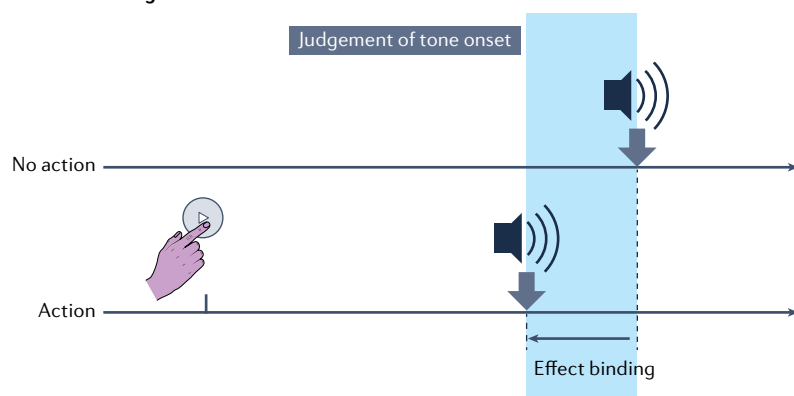
a Action binding**b Effect binding**

Fig. 1 | Measuring the intentional binding effect. Participants press a key at a time of their choosing. The action triggers a tone after a delay of 200–400 ms (typically 250 ms). **a** | Action binding is measured by the difference in the judgements of the action onset for the condition when the action does not cause a tone and for the condition when the action does cause a tone. **b** | Effect binding is measured by the difference in the judgements of the tone onset for the condition when a participant's keypress causes the tone and for the condition when the tone is not preceded by a keypress.

of agency⁴⁵. Instead, they can be useful when comparing two conditions that vary in the strength of agency⁴⁵.

Perceptual intensity. Self-induced tactile stimuli are perceived to be less intense than stimuli induced by others, a phenomenon known as sensory attenuation⁷. One example is that people usually feel low ticklishness for self-touch^{7,46} but ticklishness increases when the self-touch is delayed via a robotic arm⁴⁷. Delay diminishes the sense of agency by inducing a temporal discrepancy between an action and its effect⁴⁸. Sensory attenuation is considered to reflect weakened perceptual processing of predicted effects^{7,49,50}. However, an opposing account suggests that the perceptual processing of self-induced effects is not weakened. Instead, self-induced touch is perceived as less intense because the sensation is compared with elevated predictions of touch, known as a pre-activated baseline^{51,52}.

Owing to the link between sensory attenuation and self-initiated action, sensory attenuation is also used as an implicit measurement of the sense of agency^{53–55}. Because self-produced effects are less ‘surprising’,

measuring the degree of surprise elicited by an effect is an indirect measure of the sense of agency over that effect. Sensory attenuation is often measured using tactile stimuli and a force-matching task (FIG. 2a). In a typical paradigm, participants feel a force (pressure) and reproduce an equivalent pressure on themselves by pressing with their finger (self-touch) or by operating a pressure device (external touch). People usually apply more force in the self-touch condition than in the external touch condition^{56–58}. Sensory attenuation can be calculated by the ratio between (or the difference between) the reproduced forces of the external touch and the self-touch^{56–58}. Sensory attenuation has also been reported in the visual^{59–62} and auditory^{63–65} domains. However, sensory attenuation in the visual domain is much rarer and more ambiguous than in the tactile and auditory domains⁵⁹.

In addition to subjective ratings or intensity matching, recordings of electrophysiological event-related potentials (ERPs) are often used to measure sensory attenuation (FIG. 2b). In a typical sensory attenuation paradigm using ERP, a stimulus such as a tone, touch, or visual stimulus is either triggered by participants’ voluntary actions or by someone else. Self-triggered stimuli resulted in smaller ERPs than other-triggered stimuli^{62,64}. Furthermore, several other aspects of ERPs, such as potential latency, lateralization, and complexity, are related to sensory attenuation in voluntary actions^{62,66}. For example, one study reported that visual sensory attenuation was linked to an increase of the lateralized readiness potential⁶².

There is a loss of sensory attenuation in some patients with schizophrenia^{67,68}. However, the magnitude of sensory attenuation does not always correspond to the subjective judgement of agency in healthy participants³³. This discrepancy might be because sensory attenuation is an effect of the sense of agency on sensory feedback, whereas the judgement of agency depends on more than just sensory feedback. For example, sensory attenuation might be a result of an early sensorimotor process, but the judgement of agency can be influenced by late inference of the causal relationship between one’s action and the event. Thus, the two effects can dissociate in certain situations.

Visual attention. The sense of agency is a feeling accompanied by control over objects or events. In an environment in which control of objects and events is ambiguous, attention is automatically attracted to things that are under control^{5,69,70}. For example, when participants viewed multiple avatars and were asked to detect the avatar whose motion was identical to their own body movements, the reaction time required to detect the target avatar was very short regardless of the number of distractors³. However, when participants’ bodies were passively moved by strings or when the target avatar’s movements were temporally delayed, the search reaction time significantly increased as the number of distractors increased³. This phenomenon shows that the sense of agency can be a very salient attraction for allocation of visual attention^{3,4}.

Direct neural measures of attention including steady-state visual evoked potential (SSVEP)⁵² are used to examine visual attention directed towards controllable

Pre-activated

Neurons responsible for processing predicted sensory feedback are activated prior to the input of the actual sensory feedback, resulting in a smaller change in the activation of these neurons after the sensory feedback.

Steady-state visual evoked potential (SSVEP).

A measure of brain activity synchronized to a visual stimulus presented at a certain frequency, with an amplitude that is greatly influenced by attention.

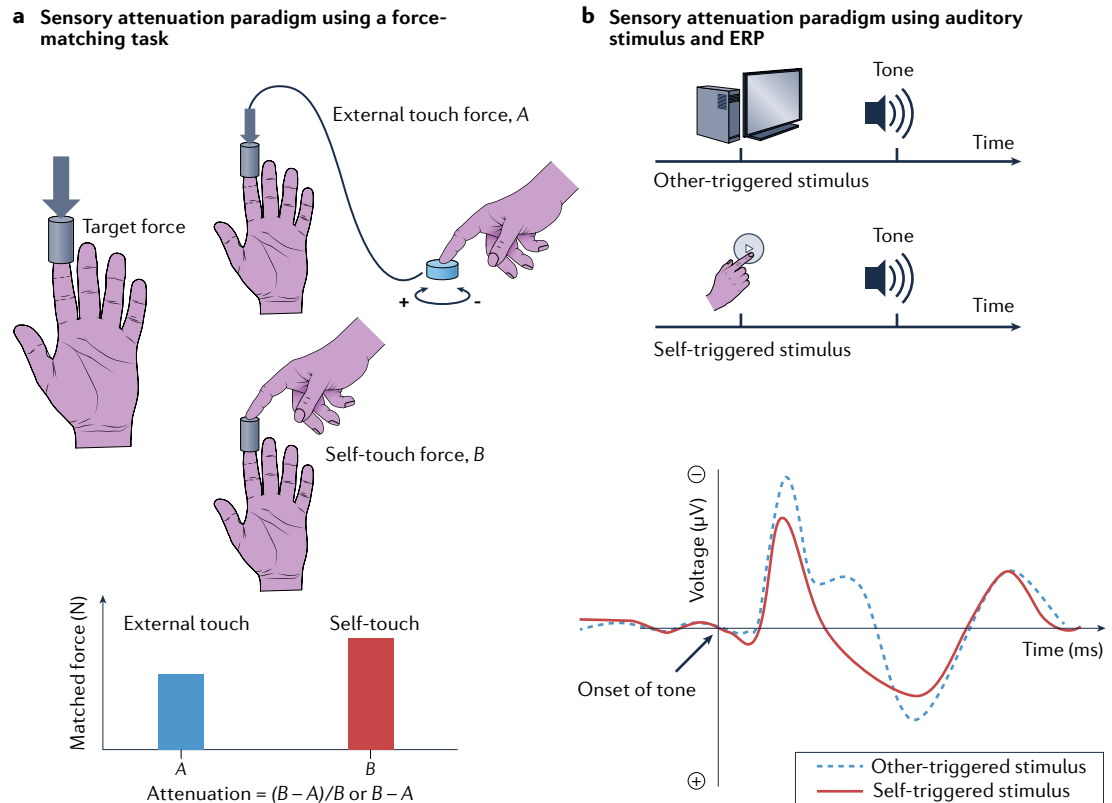


Fig. 2 | Measuring sensory attenuation. Tactile and auditory stimuli are often used to measure sensory attenuation. **a** | In a force-matching paradigm, participants have to reproduce a force that they received on their finger. People usually produce a larger force than the received force in the self-touch condition because the tactile sensation is attenuated during self-touch. The ratio between (or the difference between) the external force and the self-touch force is used as the index of sensory attenuation. **b** | In a sensory attenuation paradigm using event-related potentials (ERP), self-triggered stimuli are usually associated with smaller ERPs than other-triggered stimuli. These graphs are examples only and not linked to any actual data.

and uncontrollable objects⁷¹. In one study, participants were asked to detect a brief flash of one group of objects while continuously moving their fingers on a touchpad. The movement of one group of objects was linked to the participants' finger movements, but the movement of the other objects was not. The amplitude of the SSVEP was larger for the objects that were under control compared with the objects that were not under control, even when the task did not require the participants to pay attention to control. This finding suggests that control automatically attracts visual attention³².

Furthermore, attention paid to an object that is under control improves the visual detection of that object. In one study, participants were asked to detect a particular visual target (a circle with a single gap) among multiple distractors (circles with double gaps)⁵ (FIG. 3a). The motion of the target was a mixture of participants' real-time movements and others' movement (with a range of 0–100% participant movement)^{5,52,69,70}. The movement of the distractors was uncontrolled. There was a beneficial effect of control on visual detection such that the detection time was shorter when the participant's control over the target was larger (FIG. 3a). Thus, visual attention can be used as an indirect measure of the sense of agency⁵. A related control detection task requires participants to detect a controllable target among

multiple uncontrollable distractors^{70,72,73} (FIG. 3b). This type of task uses the discrimination of control as an index of the sense of agency and might be useful for investigating the sense of agency in ambiguous control environments both for healthy participants and for individuals whose decision-making processes are impaired⁷³.

The amplitude of SSVEPs is modulated mainly by top-down selective attention⁷¹. By contrast, studies using ERPs, which are considered to be neural indexes of early bottom-up attention, have reported an inhibitory effect of agency. Specifically, ERPs such as the N1, P3 and mismatch negativity are often observed in oddball paradigms, in which rare salient events occur among repeated stimuli. Rare events trigger larger ERP component amplitudes than do repeated events. Furthermore, these components are suppressed in the condition in which an event is self-triggered by a participant (agency condition) compared with the condition when the event is triggered by a computer (non-agency condition)^{74–79}. These phenomena can be interpreted as larger attention-related components that reflect the attention triggered by the loss of sense of agency; stimuli in agency conditions are more predictable than stimuli in non-agency conditions^{62,64}. In such cases, the loss of sense of agency automatically attracts attention, producing a larger ERP amplitude.

N1

Also known as the N100. A negative-going electrophysiological potential that peaks around 80–120 ms after the onset of an event. The peak's amplitude depends on the salience of the event.

P3

Also known as the P300. A positive-going electrophysiological potential that peaks roughly 250–500 ms after the onset of an event. The peak reflects categorization and evaluation.

Mismatch negativity

A negative electrophysiological component that peaks around 100–250 ms after the onset of a rare stimulus in a sequence of repetitive stimuli, independently of attention.

Thus, control affects attention via the sense of agency. Whether loss or gain of the sense of agency attracts attention depends on whether one has already acquired control in the environment³. Specifically, when control is uncertain in an ambiguous environment, the detection of control (the rise of the sense of agency) attracts attention to controllable objects^{3,5,52}. By contrast, when the sense of agency is expected to accompany an action, such as when pressing a key to trigger a tone, the lack of sense of agency is perceived as an unexpected circumstance and triggers more attention compared to conditions that do involve the sense of agency.

In summary, the sense of agency shapes perception in multiple ways. Agency shortens the temporal perception of the interval between one's voluntary action and the effect of action. The sense of agency also modulates the intensity of feelings regarding the effects of actions. Finally, the sense of agency changes the way the world looks by selectively allocating attention to controllable and uncontrollable subjects. These effects highlight how actively interacting with the environment shapes perception.

Sense of agency influences behaviour

The sense of agency also influences human interactions with the world. For instance, the ability to recognize the consequences of one's own actions might play an important part in the development and organization of voluntary motor control⁸⁰. In this section, we discuss the effects of the sense of agency on action selection, goal-directed behaviour and social cognition.

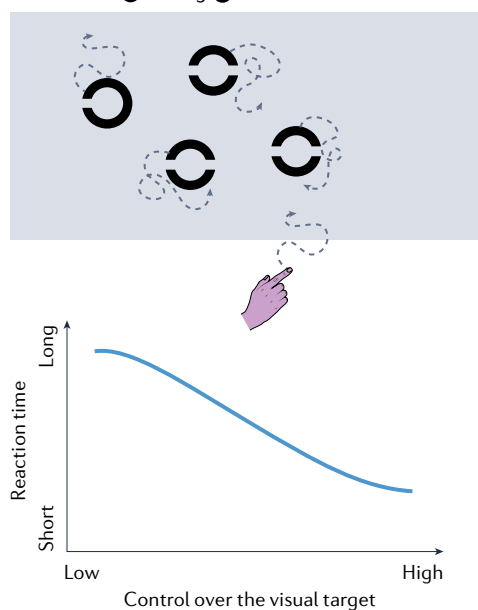
Action selection. Newborn infants have little voluntary control over their own bodies. However, signs of exploration of self-agency are exhibited from an early age. Ten-week-old infants can recognize the contingency between their leg movements and the movements of a mobile tethered to their leg, and react by increasing their leg movements⁸. Two-month-old infants generate more frequent pressure on a pacifier just at the threshold at which it produces congruent sounds⁸¹. Furthermore, 8-month-old infants can control their gaze direction to trigger visual effects^{82,83}. Thus, the exploration of causal action–effect contingency starts at a very early age and modulates infants' selection of what action to execute. This behavioural reaction to contingency is considered evidence of the early emergence of the sense of agency^{11,84}.

The sense of agency also influences action selection (what action to execute) and action execution (how to execute the action) in adults^{85,86}. In one study, participants were asked to freely press one of four keys after the appearance of a visual cue⁸⁵. The keys caused the appearance of a visual outcome with different probabilities. Keys that were more likely to result in a visual outcome were more likely to be selected and were pressed more quickly than keys that did not. Thus, actions that are associated with a better sense of agency are preferred and are executed more fluently.

The sense of agency over an action facilitates the selection of that action and also improves action fluency through an internal reward system^{85,86} (FIG. 4). The sense of agency is an internal reward that facilitates many aspects of action. This facilitation effect is sensitive to

a Reaction time in visual search

Search for a ○ among ○ distractors



b Control detection accuracy

Search for the object that is under control

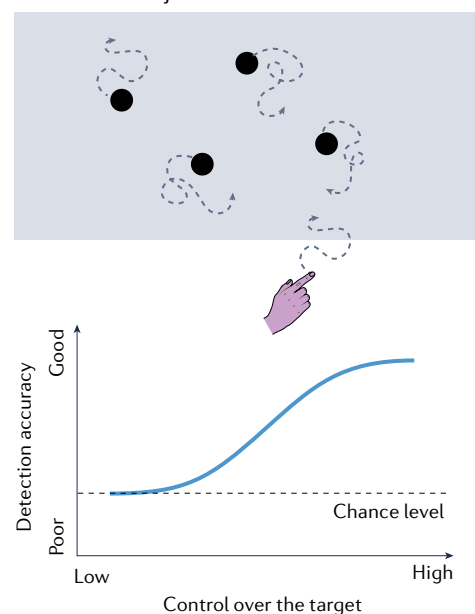


Fig. 3 | **Measuring the influence of control.** **a** | To measure the impact of control on visual attention, participants detect a visual target among multiple distractors. The target is under a certain level of control by the participant but the distractors are not. Control over the target attracts attention and shortens the reaction time for detecting the target. **b** | To measure discriminability of control, participants detect a controllable object among several uncontrollable objects that are visually identical to the target. The stronger the control over the target, the better the detection accuracy. Both paradigms can be used as indirect measurements of the sense of agency.

the effectiveness of the motor response (how likely the response is to evoke an outcome)^{87,88} but not to monetary rewards⁸⁹. Thus, the effects of internal rewards (sense of agency) and external rewards (monetary rewards) can be dissociated. The facilitation effect in adults is consistent with infant studies that showed that control reinforces action. Furthermore, action–outcome contingency (how consistently an action triggers an outcome) — which is critical to the sense of agency — has an important role in determining whether response speed is governed by the outcome probability⁹⁰. Reliable control, in which as an action consistently produces an expected outcome, is linked to action motivation. In addition, the reinforcement of actions from the sense of agency is sensitive to increased, decreased, and unchanged reliability of control⁸⁸.

Finally, the sense of agency is relevant to motor learning. In one study of the intentional binding effect, people showed a stronger action binding following a trial in which they made an error than following a trial that was correct⁹¹. Importantly, the positive inter-individual correlation between this post-error boost and the learning rate from the errors (the rate of adjusting actions to be more successful) implied that people who showed a strong post-error agency boost also learned from their errors⁹¹. Furthermore, a study reported a significant positive correlation across individuals between adaptive motor learning and the perceptual sensitivity of detecting an increase in control⁹². Taken together, these studies show that the sense of agency affects how well people adjust their behaviour according to changes in their environment⁹².

Goal-directed behaviour. Knowing whether one has control in an environment plays an important part in motivating exploratory behavior. By contrast, knowing which things one can control is critical for exploitative or goal-directed behaviour. The trade-off between exploration and exploitation in the context of ambiguous outcomes is a well-known decision-making problem⁹³. Exploration increases the possibility of acquiring new outcomes, whereas exploitation maximizes the current known outcome. Rewards such as points and monetary gains play an important part in choosing between exploration and exploitation^{94–97}: Rewards trigger goal-directed exploitative behaviour and reduce exploratory behaviour⁹⁴.

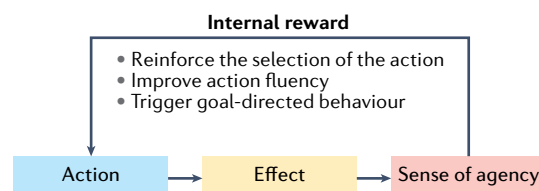


Fig. 4 | The effect of sense of agency on action as an internal reward. Action includes the intention, selection and execution of action. Effect refers to the consequence of the action. The sense of agency can modulate people's action through an internal reward mechanism, reinforcing the selection of an action which can reliably trigger effects, improving action fluency, and triggering goal-directed behaviours.

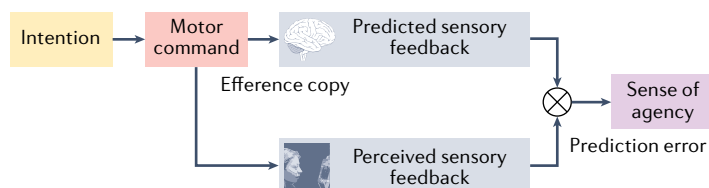
Overall, control in the environment is important for increasing action frequency, whereas control over a specific object can trigger goal-directed behaviours. For instance, when participants move a mouse to trigger movements of multiple objects on the screen and are asked to detect whether they could control one of the objects, they make fewer exploratory movements before responding in a condition in which they have better control over an object^{72,98}. This reduction in movements is considered to be linked to precise goal-directed behaviour when the control is detected⁹⁸. In another study, participants had a certain level of control over all stimuli and had to determine the exact level of control over each stimulus⁶⁹. Kinematic analysis of participants' movement frequency across time within the trial showed that the average level of control over the stimuli initially boosted the moving frequency and the detection of target control reduced the moving frequency. Together, these studies show a shift from exploratory to exploitative action associated with control and the sense of agency⁶⁹.

Social cognition. The sense of agency is also associated with an individual's sense of identity and autonomy. Specifically, when people feel that they are in control of their own actions and the consequences of those actions, they believe that they have the freedom to make decisions and take action. Such autonomic decisions and actions are fundamental to a sense of identity. Furthermore, investigation of the sense of agency in social interactions provides important insights into individual autonomy and responsibility in philosophy and ethics.

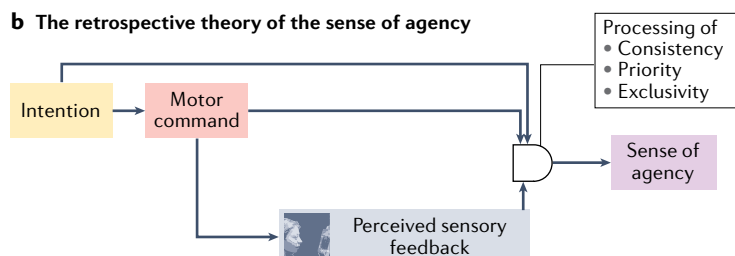
Many laws handle unintended and intended consequences differently⁹⁹ and it is debatable whether people should be responsible for their behaviour if they lack a sense of agency¹⁰⁰. For example, classic work has indicated that people perform excessive actions when they feel that they are being obedient compared with when they make decisions by themselves¹⁰¹ (but see recent criticisms and reinterpretations¹⁰²). Studies re-examining this research show that obedience decreases the sense of agency over one's own actions and reduces neural processing of the outcomes resulting from one's actions^{100,103}. Furthermore, both 'commanders' (who gave orders) and 'agents' (who received orders and performed actions) exhibited a reduced sense of agency when in these roles compared with when they had autonomy and could act independently¹⁰⁴. The mere presence of other agents can also reduce the sense of agency^{105,106}. In summary, the sense of agency is not limited to one's own behaviour; it is also influenced by other agents and can influence interactions among people.

Virtual reality has been used to investigate how sense of agency over a body influences social cognition. In virtual reality, when people control a virtual body via a first-person perspective they acquire a sense of ownership over that body¹⁰⁷. A sense of agency over the virtual body, based on the congruence between participants' movements and viewed movements of the virtual body, has an important role in acquiring and maintaining this sense of ownership^{108–111}. Controlling a virtual body that is different from one's real body can alter social

a The comparator model of the sense of agency



b The retrospective theory of the sense of agency



c The Bayesian integration model

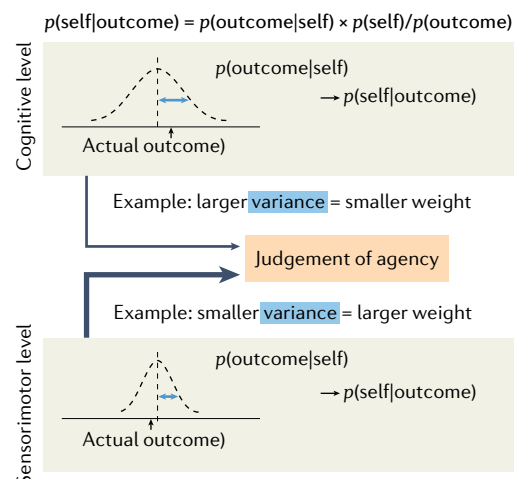


Fig. 5 | Theories of the sense of agency. **a** | The comparator model suggests that the sense of agency is generated by the comparison between predicted sensory feedback and perceived sensory feedback. **b** | The retrospective theory does not assume a prediction in the brain, but instead that one feels a sense of agency when the outcome and the action/intention meet the consistency, priority and exclusivity principles. **c** | The Bayesian integration framework posits that the probabilities of self-agency at the sensorimotor and cognitive levels are calculated from the prior distribution of outcome and actual input, weighted by the variabilities of the prior distributions. Probabilities at the two levels are integrated into a judgement of agency.

cognition, such as implicit racial discrimination^{112,113} and acting style^{114,115}. For instance, white participants' implicit racial bias against Black people was reduced after they had embodied the virtual body of a Black person¹¹³. In a similar vein, adult participants' estimation of object size significantly increased after they controlled the virtual body of a 4-year-old child¹¹⁵. These studies demonstrate that people acquire a sense of identity through the sense of agency, and their sense of identity in turn shapes the way they interact with the world.

In summary, the sense of agency substantially shapes human behaviour. Perceiving control in the environment reinforces exploratory behaviours and detecting a high level of control triggers exploitative behaviours to maximize rewards. Furthermore, the sense of agency affects one's social role and sense of identity. These findings provide clues to how humans' sense of agency modifies their interactions with their environment.

How sense of agency arises

Despite the importance of the sense of agency, it is unclear exactly how it arises. For example, it is unknown whether a baby who shows signs of noticing the relationship between their leg movements and the movements of a mobile feels a sense of agency. It is likewise unclear why some patients with schizophrenia report no sense of agency over their actions. In this section, we introduce theories of the sense of agency and neuroimaging findings that attempt to clarify the cognitive and neural bases of agency.

Theories of the sense of agency. The most widely accepted theory of the sense of agency is the comparator model^{7,116,117} (FIG. 5a), which contains two key components: a sensory prediction and a comparison mechanism.

The theory states that when people execute a voluntary action, a sensory prediction based on an efference copy of the motor command is generated in the brain and compared with the actual sensory feedback received from the sensory system and the external world^{7,116,117}. A sense of agency arises if the prediction and the sensory feedback are congruent and diminishes if there are large mismatches (also known as prediction error or sensorimotor incongruity). For example, when intending to pick up a water bottle, the brain sends motor commands to the arm muscles to perform the action. The efference copies of the motor commands generate sensory predictions, such as how the water bottle will feel on the skin and changes in proprioception. After picking up the bottle, the predicted sensory feedback is compared with the actual tactile and proprioceptive feedback. If the comparison reveals no sensorimotor incongruity, a sense of agency arises.

This account is well supported: empirical studies show that sensorimotor incongruity greatly reduces the sense of agency^{31,48,118–120}. For example, participants experienced a lack of sense of agency when their action produced an unpredicted tone^{31,32,118}. Furthermore, participants experienced an illusory sense of agency when externally generated events incidentally matched their predictions^{118,121}. However, there are some issues with the comparator model. For example, people do not always generate predictions regarding what would change in the external world for exploratory actions, but can still perceive the sense of agency after sensing a regular pattern between their action and feedback⁷².

By contrast, retrospective theory suggests that the sense of agency is experienced from the cognitive inference that one's thoughts have caused one's actions^{121,122} (FIG. 5b). The sense of agency can be based on a causal

Efference copy

The internal duplicates of a motor command in the brain, used to predict the sensory feedback of the motor command.

Proprioception

The internally generated sense of self-movement and body position.

inference when three principles are met: actions are consistent with the outcomes (consistency), actions are conducted before the outcomes (priority) and there are no alternative causes of the outcomes (exclusivity)^{121,122}. Under this framework, a prediction is no longer necessary, and explorative actions that lack certain predictions can still generate a sense of agency as long as the above principles are satisfied.

These are multiple aspects of the sense of agency that are not considered in the comparator model or retrospective theory. Many factors, including task performance^{123–127}, mental state^{128,129} and social interactions^{105,106}, influence the sense of agency. For example, participants report a weakened sense of agency when another agent is present, even when the other agent has no control over the outcome of a task^{105,106}. These effects are not consistent with either theory.

Bayesian integration provides a new computational framework that can incorporate additional factors relevant to the sense of agency. Integration theory posits that the sense of agency arises from a Bayesian computation over two levels: sensorimotor and cognitive (FIG. 5c). The sensorimotor level contains the factors that are directly linked to actions and action outcomes, and the cognitive level contains factors that are not directly linked to actions and action outcomes but instead are linked

to intentions, expectation and inferences. The probability that a sense of agency will be felt can be calculated from the prior distribution of outcomes and the actual input of outcomes by applying Bayes' rule at each level, weighting by the variabilities of the outcome distributions^{130,131}. This weighting principle is similar to the statistical optimal weighting of visual and haptic information in multisensory integration¹³². Bayesian integration theory can explain the effects of many factors that are not directly linked to actions but that influence the prior distribution of outcomes (BOX 1).

Theories provide frameworks to understand how the sense of agency (or its breakdown) can be calculated from internal and external inputs. Studies on the neural basis of the sense of agency might provide further important insights into the questions of how the sense of agency arises and how it relates to other cognitive systems such as decision-making, causal inference, and reinforcement learning¹³³.

Neural basis of the sense of agency. Functional magnetic resonance imaging (fMRI) has been used to identify brain regions associated with the sense of agency. The comparator model in particular has prompted the study of regions related to sensorimotor incongruency, induced (using manipulations such as delayed effects and unexpected events) to disturb the sense of agency. Researchers compare brain activity between congruent (strong agency) and incongruent (weak agency) conditions^{45,134}. For example, in a congruent condition, the direction of a cursor movement matches the direction in which participants move their joysticks, whereas in the incongruent condition the cursor moves in the opposite direction¹³⁵. Studies have often found stronger brain activity in frontal-parietal regions^{136–138}, such as the angular gyrus in the inferior parietal lobe^{45,135,139–141}, pre-supplementary motor area^{139,142}, and the temporo-parietal junction^{143,144} in incongruent conditions compared to congruent conditions. By contrast, other studies have highlighted brain regions where activity increases with the detection of congruency (compared to a condition in which no congruency is detected), such as the insula^{135,136,139} and precuneus^{145,146}. However, it remains unclear whether there is a central region where the sense of agency arises directly on the basis of action–feedback congruency across the above-mentioned regions.

The sense of agency has also been decoded from fMRI activity patterns¹⁴⁴. One study found that the right supramarginal gyrus (SMG) and left inferior frontal gyrus were sensitive to subjective (explicit) judgements of the sense of agency. By contrast, the bilateral precentral gyri and the left inferior parietal lobe (IPL) were sensitive to prediction error (sensorimotor incongruency). The decoding accuracy for explicit agency reached its maximum value in the right SMG a few seconds after prediction error could be successfully decoded in the precentral gyri and the left IPL, which suggests that the right SMG represents explicit agency based on prediction error information from other regions. These data suggest the importance of the right IPL, including the angular gyrus and the SMG, to producing an explicit sense of agency^{139,147}. The left IPL has been found to be

Box 1 | The Bayesian integration model

Bayesian theory states that the probability of an event is based on prior knowledge of the conditions related to the event. For the sense of agency, the probability of whether an event is the result of one's own action depends on the prior distribution of events that could be caused by the action. The Bayesian formula for the judgement of agency from a perceived outcome can be written as follows:

$$p(\text{self}|\text{outcome}) = p(\text{outcome}|\text{self}) \times p(\text{self})/p(\text{outcome})$$

$p(A|B)$ refers to the probability of A occurring given that B is true, and $p(A)$ refers to the probability of observing A without any given conditions. $p(\text{self}|\text{outcome})$ refers to the probability that the self perceives a sense of agency given the outcome; $p(\text{outcome}|\text{self})$ refers to the probability that the self could cause the outcome; $p(\text{self})$ refers to the overall probability of the self as an agent in the environment; and $p(\text{outcome})$ refers to the probability of the outcome occurring in the environment.

This framework aids in understanding some specific findings about the sense of agency. For example, delayed feedback greatly impairs the sense of agency⁴⁸. However, the length of delay sufficient to diminish the sense of agency varies among different circumstances⁴⁸, and there are large individual differences in the sense of agency for delayed events^{33,48}. Neither the comparator model nor the retrospective theory can address this uncertainty. According to the Bayesian account, it can be described by the prior distribution of the outcome ($p(\text{outcome}|\text{self})$). Different actions, such as turning on a light and turning on the shower tap, are associated with different prior distributions of delay in feedback. One usually expects less than half a second of delay in the light, but might expect a delay of a few seconds for water to come out of the tap. Furthermore, in experimental tasks, the prior distribution of delay can be greatly influenced by individual differences and experimental conditions, resulting in large variability in the judgement of agency.

Furthermore, this framework also separates the outcome distribution into sensorimotor and cognitive levels. The prior distribution of the outcome at the sensorimotor level contains the distribution of the direct action outcomes. The prior distribution of outcomes at the cognitive level contains expectations of the outcomes. For example, the presence of another agent reduces the sense of agency even when the other agent does not influence any events¹⁰⁵, which can be explained by the prior assumption that other agents can potentially produce events. This assumption decreases both the probability of a certain outcome in the case of the self ($p(\text{outcome}|\text{self})$) and the probability of self ($p(\text{self})$), resulting in a decrease in the probability of the self ($p(\text{self}|\text{outcome})$).

Cerebellar forward models

Motor control models, acquired in the cerebellum, that transform a set of efference copies of motor commands into predicted sensory feedback.

Electromyogram

Recording of electrical activity produced by skeletal muscles from the skin surface above the muscle.

sensitive to sensorimotor information (prediction error) rather than explicit agency, which is consistent with the finding that anodal electric stimulation to the left parietal cortex diminishes intentional binding¹⁴⁸.

In addition to the cortical regions discussed above, the cerebellum and basal ganglia have critical roles in predicting action outcomes, including sensory feedback¹⁴⁹ and reward¹⁵⁰. Cerebellar forward models^{151–154} constitute an essential part of the comparator model and many studies support the importance of a cerebellum–cortex loop in sensory attenuation^{149,155,156}. By contrast, little is known about the contribution of the basal ganglia to the sense of agency, although one study reported weaker functional connectivity between the SMG and basal ganglia in patients with schizophrenia with impairments in the sense of agency¹⁵⁷. Furthermore, dopamine has a vital role in the functions of the basal ganglia and dopaminergic medication boosts the sense of agency in patients with Parkinson's disease¹⁵⁸. Future studies on the roles of the basal ganglia and the dopamine system will enhance understanding of the neural mechanisms underlying how the sense of agency facilitates motivation, internal reward and exploration.

Returning to theoretical accounts of the sense of agency, the comparator model has received support from studies that reported increased brain activity in frontal-parietal regions when mismatches between motor predictions and sensory feedback are detected^{136–138}. By contrast, little neural evidence has been accumulated for the retrospective theory, probably owing to the difficulty in experimentally controlling the cognitive factors affecting causal inference. Finally, there is also a lack of neural data regarding the integrative model. It remains unclear how the probability of feeling a sense of agency at the cognitive and sensorimotor levels might be integrated in the brain.

Agency in human–machine interactions

One aim of studying the sense of agency is to understand the properties and mechanisms of the phenomenon. A more applied aim is to understand how the sense of agency affects real-life situations in which humans interact with other humans or with machines. Human–machine interaction technologies have been rapidly developing in the past few decades and their use has the potential to influence how humans feel a sense of agency. For example, many new vehicle models are equipped with driving assistance functions such as cruise control, lane-keeping assistance and automatic emergency braking¹⁵⁹. In another domain, semi-automated remote control systems have been developed to assist humans in precise operations such as surgery^{160,161}. In both examples, human control is still dominant and automated systems take partial control to assist and reduce human error. In human–machine joint control situations, understanding how people feel a sense of agency and the resulting changes to behaviour is critical to the design of safe, efficient, and human-friendly systems^{159,162}.

To understand human–machine joint control, it is instructive to consider human–human joint control. In joint control with two human agents, the presence of another agent usually reduces the sense of agency because

it weakens each agent's actual control^{163–165}. The actions of the other agent are often unpredictable and produce sensorimotor discrepancies from one's own predictions, resulting in a lower sense of agency compared with solo control. In human–machine joint control, a reduction in the sense of agency has been reported in driving automation¹⁶⁶ and remote control of construction machines¹⁶⁷. The decrease in sense of agency in human–machine joint control is also due to the sensorimotor discrepancy introduced by control by the automated system.

Decreased sense of agency can result in poor human engagement in joint human–machine control. For example, one study tested participants' ability to avoid crashes in a driving simulation with machine automation. The automation precisely followed the road and maintained speed. Participants were instructed to supervise the driving at all times and could override the automation by steering or braking at any time¹⁶⁸. Although a reminder system eliminated long eye-off-path durations that would otherwise contribute to inattentiveness, 28% of participants failed to avoid crashing after driving for 30 min with driving automation¹⁶⁸. The study design ensured that drivers were looking at the road, so driver engagement was concluded to have a cognitive component rather than being purely visual¹⁶⁸. The sense of agency might be a critical factor underlying cognitive engagement in such a joint control situation¹⁵⁹.

Cognitive studies on the sense of agency suggest that a 'we-mode' might be critical to maintain a sense of agency in both human–human and human–machine joint control¹⁶³. We-mode refers to when multiple agents share attention, perspectives, and intentions in a task¹⁶⁹. For example, a we-mode can be established when two people carry a heavy object together and communicate their goal to walk in a certain direction. To date, we-mode is mainly discussed in human–human interactions, and little is known about establishing a we-mode in human–machine joint control.

Some methods have been proposed to decode human intention prior to action using measures such as an electromyogram, eye movements and electroencephalography^{170–173}. For example, when a human carries a heavy object together with a robot arm, electromyogram signals measured from the human's upper limb can be used to predict the human's motor intention and convey their intention to the robot¹⁷¹. Sharing human intention with the automated system helps to maintain a sense of agency in human–machine joint control and to improve human task performance¹⁷⁴. For example, a previous study showed that tactile and proprioceptive information enables a human to estimate their robot partner's movement goal to improve their own motor performance¹⁷⁵. In a driving assistance study, when human drivers intended to decelerate their vehicles and the systems conducted an earlier and smoother deceleration than the drivers' actual actions, the sense of agency was maintained at a level comparable to when the driver had full control over the vehicle, and driving performance was significantly improved by the deceleration assistance¹⁷⁴. In line with this perspective, a study on human–machine cooperative control proposed a framework of driving assistance to provide the 'best response'

for human drivers, in which humans decide what to do, communicate their intention to the system, and the system then helps them to perform optimally¹⁷⁶.

In instances when input at the sensorimotor level is relatively less reliable owing to limited visual input and time lags — such as in teleoperation and virtual reliability — cues at the cognitive level might have a large impact on the sense of agency¹²³. In such cases, the sense of agency may rely less on whether the sensory feedback corresponds to one's motion and more on whether the outcome matches one's expectation. Thus, interventions at the cognitive level, such as presenting modified outcomes that are slightly more optimal than one's actual movement, might even enhance the sense of agency in human-machine joint control compared with solo control^{123,125}. Some evidence in favour of this hypothesis comes from a visual modification in virtual reality that reduced perceived motion error by half and improved the sense of agency over a moving object, even when the object did not reflect the exact position of the participant's hand¹²⁵.

In summary, sharing intentions between systems and humans and combining machine and human control might be useful to maintain or even enhance the sense of agency in human-machine interactions. Finally, although an artificial agent usually has less impact on social interaction than a real human agent, communicating the system's intention with human users is another useful approach to establishing a we-mode and to preventing a reduction in the sense of agency in human-machine joint control.

Summary and future directions

The sense of agency is a hub between action, perception, and consciousness and is critical to how humans perceive and interact with the world. The sense of agency influences time perception of actions and their consequences, sensation, attention allocation, action selection, and goal-directed behaviour. Furthermore, feelings about the sense of agency are influenced by and can alter human-machine interactions. However, the answers to many research questions associated with the sense of agency remain unknown.

More work is needed to develop reliable methods for measuring the sense of agency through behaviour. Changes in time perception, kinematics, and action selection are useful for assessing the sense of agency indirectly. It is also worth discussing how processes underlying the sense of agency are implemented in the cognitive system and how they influence and overlap with other processes to influence motor control¹⁷⁷.

A computational model of the sense of agency is not yet well established and verified. Although a Bayesian approach provides a useful computational framework, experiments that manipulate the input and reliability of each cue and that can assess the sense of agency under different conditions are required to further develop such a model. A computational model of the sense of agency could be used to estimate the sense of agency under different circumstances and to simulate changes in the sense of agency in human-machine interaction when designing new interfaces.

In the field of psychiatry, patients with schizophrenia show an abnormal sense of agency when some components underlying this sense produce inaccurate output or the integration of multiple cues is unbalanced. Diagnosis and treatment studies are needed to determine how and why disorders in the sense of agency occur. Examining disorders in the prior distribution of outcomes and disorders in Bayesian inference might be a useful approach. Furthermore, once a computational model of the sense of agency has been established, it might be possible to simulate the impaired sense of agency and the possible effects of treatments.

Finally, the fundamental biological question of how electrochemical neuronal activity in the brain generates the sense of agency is linked to the 'hard problem of consciousness' — how physical neural activity forms the subjective experience of consciousness — and remains largely unknown. Future neuroscience research should identify not only the brain regions that are responsible for the sense of agency, but also the processing route of this subjective feeling.

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